

STD – X
Subject : Science



CHAPTER 1
**Chemical Reactions
and Equations**





CHAPTER 1

Chemical Reactions and Equations

Part – 6

Types of Chemical Reactions

D) Double Displacement Reaction



D) Double Displacement Reaction

- A chemical reaction in which the positive ions (cation) and the negative ions (anion) of the two reactants (exchange) with each other, (forming precipitate) is called double displacement reaction.



A and C are Cations (Positive Ions)
B and D are Anions (Negative Ions)

Double Displacement Reaction

4) Double Displacement Reaction

Activity 1.10 : Sodium Sulphate and Barium Chloride

Activity 1.10

- Take about 3 mL of sodium sulphate solution in a test tube.
- In another test tube, take about 3 mL of barium chloride solution.
- Mix the two solutions (Fig. 1.9).
- What do you observe?

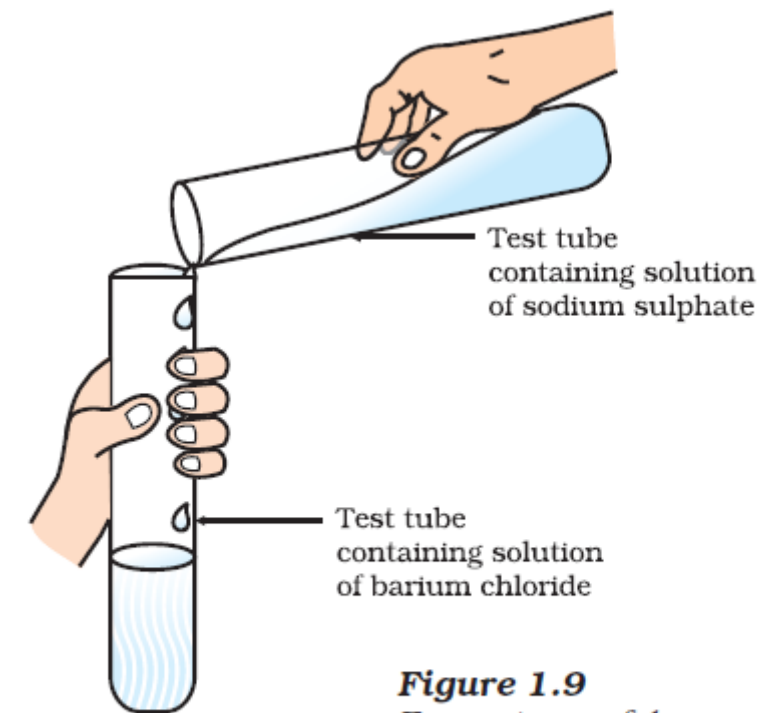


Figure 1.9
Formation of barium sulphate and sodium chloride

action.

(1.27)

4) Double Displacement Reaction

Activity 1.10 : Sodium Sulphate and Barium Chloride

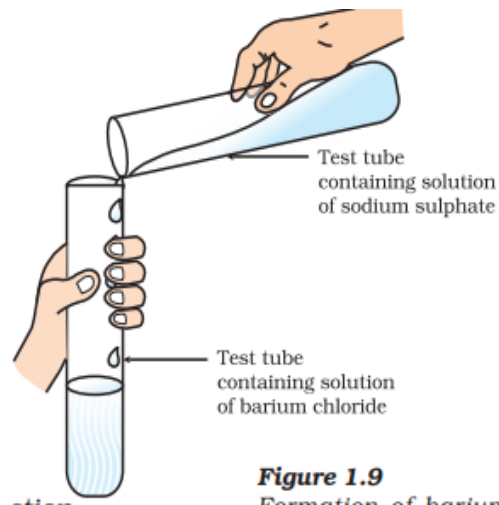


Figure 1.9
Formation of barium



Sodium
sulphate

Barium
Chloride

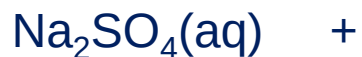
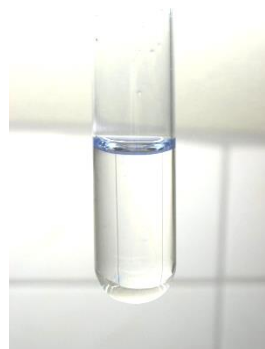
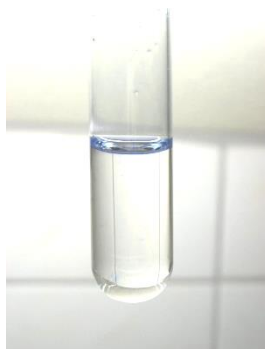


Barium
Sulphate
(White Precipitate)

Sodium
Chloride

4) Double Displacement Reaction

- Activity – 1.10 : Sodium Sulphate and Barium Chloride



+



→



+



Sodium
sulphate

Barium
Chloride

Barium
Sulphate
(White PPT)

Sodium
Chloride

- Observations :**
- Two ions Sodium (Na⁺) and Barium (Ba⁺) exchange with each other. This is a 'double displacement reaction'.
- White precipitate of BaSO₄ is formed by reaction of Ba²⁺ and SO₄²⁻. Thus, it is also precipitation reaction.
- Sodium chloride (NaCl) remains in the solution.
- Conclusion / Inference :**
- The reaction between Sodium sulphate and Barium chloride to form a white precipitate is a 'double displacement reaction.'

4) Double Displacement Reaction

Activity : 1.2 – Reaction between Lead Nitrate and Potassium Iodide .



Lead
Nitrate

+



Potassium
Iodide



Lead
Iodide
(Yellow ppt)

+



Potassium
Nitrate

Activity 1.2

- Take lead nitrate solution in a test tube.
- Add potassium iodide solution to this.
- What do you observe?

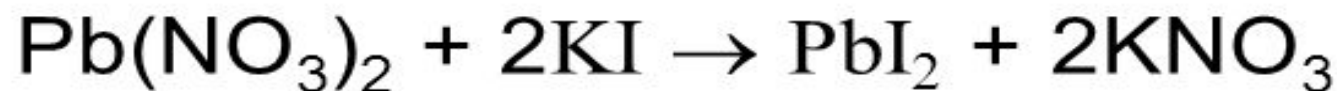
4) Double Displacement Reaction

Activity – 1.2 : Lead nitrate and potassium iodide solution.

Observations :

When potassium iodide solution is poured into a solution of lead(II) nitrate, a double-replacement reaction takes place.

Lead(II) iodide forms as a yellow precipitate. Potassium nitrate remains in solution.



Conclusion :

The reaction between 'lead nitrate' and 'potassium iodide' to form a yellow precipitate is a 'double displacement reaction.'

TO BE CONTINUED

